Database Systems: Design, Implementation, and Management

Eighth Edition

Chapter 7

Introduction to Structured Query Language (SQL)

Objectives

- In this chapter, you will learn:
  - The basic commands and functions of SQL
  - How to use SQL for data administration (to create tables, indexes, and views)
  - How to use SQL for data manipulation (to add, modify, delete, and retrieve data)
  - How to use SQL to query a database for useful information

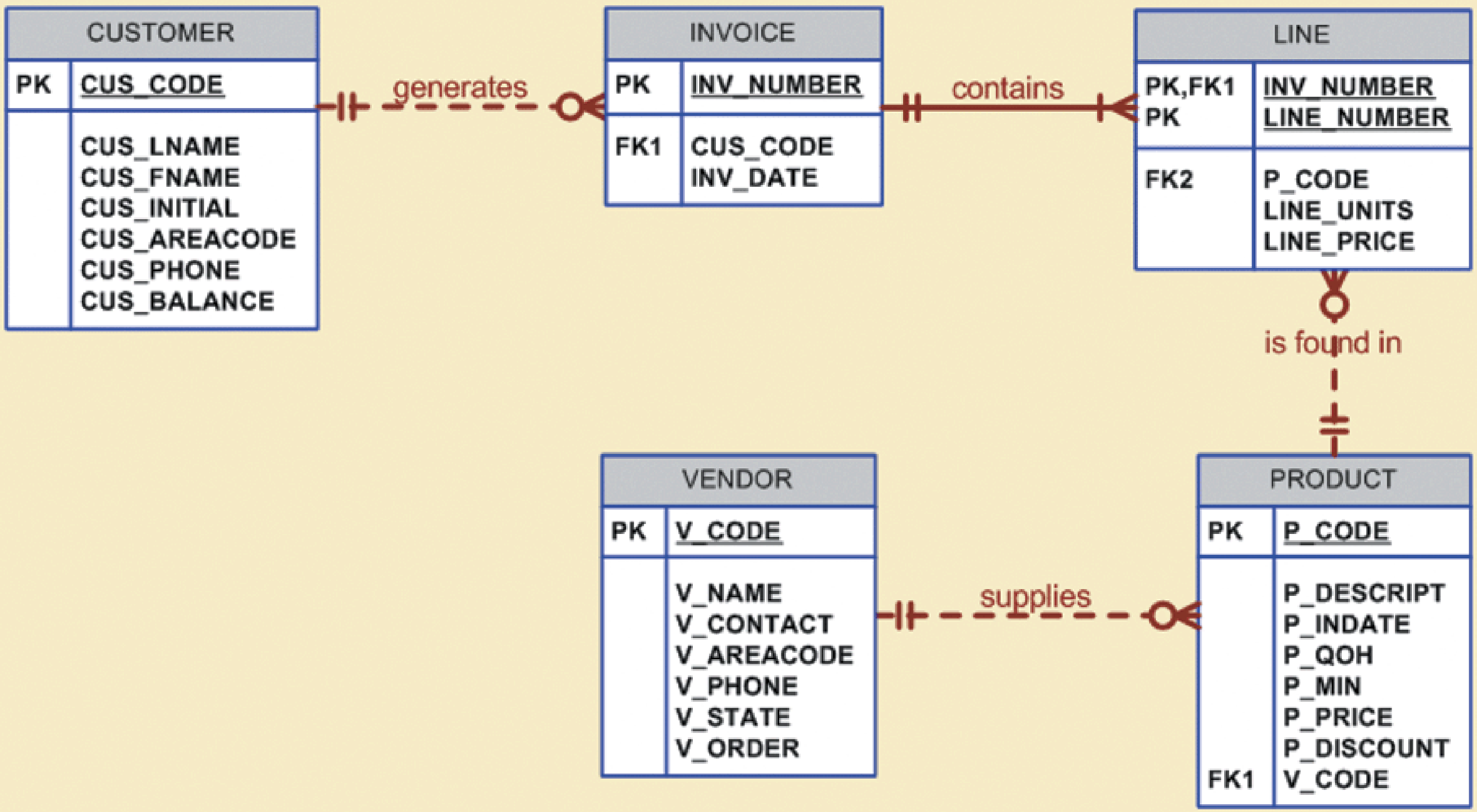
Introduction to SQL

- SQL functions fit into two broad categories:
  - Data definition language
  - Data manipulation language
- Basic command set has vocabulary of less than 100 words
- American National Standards Institute (ANSI) prescribes a standard SQL
- Several SQL dialects exist

Data Definition Commands

- The database model
  - In this chapter, a simple database with these tables is used to illustrate commands:
    - CUSTOMER
    - INVOICE
    - LINE
    - PRODUCT
    - VENDOR
  - Focus on PRODUCT and VENDOR tables

FIGURE 7.1 The database model



Creating the Database

- Two tasks must be completed:
  - Create database structure
  - Create tables that will hold end-user data
- First task:
  - RDBMS creates physical files that will hold database
  - Differs substantially from one RDBMS to another

The Database Schema

- Authentication
  - DBMS verifies that only registered users are able to access database
  - Log on to RDBMS using user ID and password created by database administrator
- Schema
  - Group of database objects that are related to each other

Data Types

- Data type selection is usually dictated by nature of data and by intended use
- Supported data types:
  - Number(L,D), Integer, Smallint, Decimal(L,D)
  - Char(L), Varchar(L), Varchar2(L)
  - Date, Time, Timestamp
  - Real, Double, Float
  - Interval day to hour
  - Many other types

Creating Table Structures

- Use one line per column (attribute) definition
- Use spaces to line up attribute characteristics and constraints
- Table and attribute names are capitalized
- NOT NULL specification
- UNIQUE specification

Creating Table Structures (continued)

- Primary key attributes contain both a NOT NULL and a UNIQUE specification
- RDBMS will automatically enforce referential integrity for foreign keys
- Command sequence ends with semicolon

SQL Constraints

- NOT NULL constraint
  - Ensures that column does not accept nulls
- UNIQUE constraint
  - Ensures that all values in column are unique
- DEFAULT constraint
  - Assigns value to attribute when a new row is added to table
- CHECK constraint
  - Validates data when attribute value is entered

SQL Indexes

- When primary key is declared, DBMS automatically creates unique index
- Often need additional indexes
- Using CREATE INDEX command, SQL indexes can be created on basis of any selected attribute
- Composite index
  - Index based on two or more attributes
  - Often used to prevent data duplication

Data Manipulation Commands

- INSERT
- SELECT
- COMMIT
- UPDATE
- ROLLBACK
- DELETE

Adding Table Rows

- INSERT
  - Used to enter data into table
  - Syntax:
    - INSERT INTO *columnname*
      VALUES (*value1*, *value2*, … , *valueN*);

Adding Table Rows (continued)

- When entering values, notice that:
  - Row contents are entered between parentheses
  - Character and date values are entered between apostrophes
  - Numerical entries are not enclosed in apostrophes
  - Attribute entries are separated by commas
  - A value is required for each column
- Use NULL for unknown values

Saving Table Changes

- Changes made to table contents are not physically saved on disk until:
  - Database is closed
  - Program is closed
  - COMMIT command is used
- Syntax:
  - COMMIT [WORK];
- Will permanently save any changes made to any table in the database

Listing Table Rows

- SELECT
  - Used to list contents of table
  - Syntax:
    - SELECT *columnlist*
    - FROM *tablename*;
- Columnlist represents one or more attributes, separated by commas
- Asterisk can be used as **wildcard** character to list all attributes

Updating Table Rows

- UPDATE
  - Modify data in a table
  - Syntax:

    UPDATE *tablename*
    SET *columnname* = *expression* [, *columnname* = *expression*]
    [WHERE *conditionlist*];
- If more than one attribute is to be updated in row, separate corrections with commas

Restoring Table Contents

- ROLLBACK
  - Undoes changes since last COMMIT
  - Brings data back to pre-change values
- Syntax:
  - ROLLBACK;
- COMMIT and ROLLBACK only work with commands to add, modify, or delete table rows

Deleting Table Rows

- DELETE
  - Deletes a table row
  - Syntax:

    DELETE FROM *tablename*
    [WHERE *conditionlist*];
- WHERE condition is optional
- If WHERE condition is not specified, all rows from specified table will be deleted

Inserting Table Rows with a SELECT Subquery

- INSERT
  - Inserts multiple rows from another table (source)
  - Uses SELECT subquery
  - **Subquery**: query embedded (or nested) inside another query
  - Subquery executed first
  - Syntax:

    INSERT INTO *tablename* SELECT *columnlist*
    FROM *tablename*;

SELECT Queries

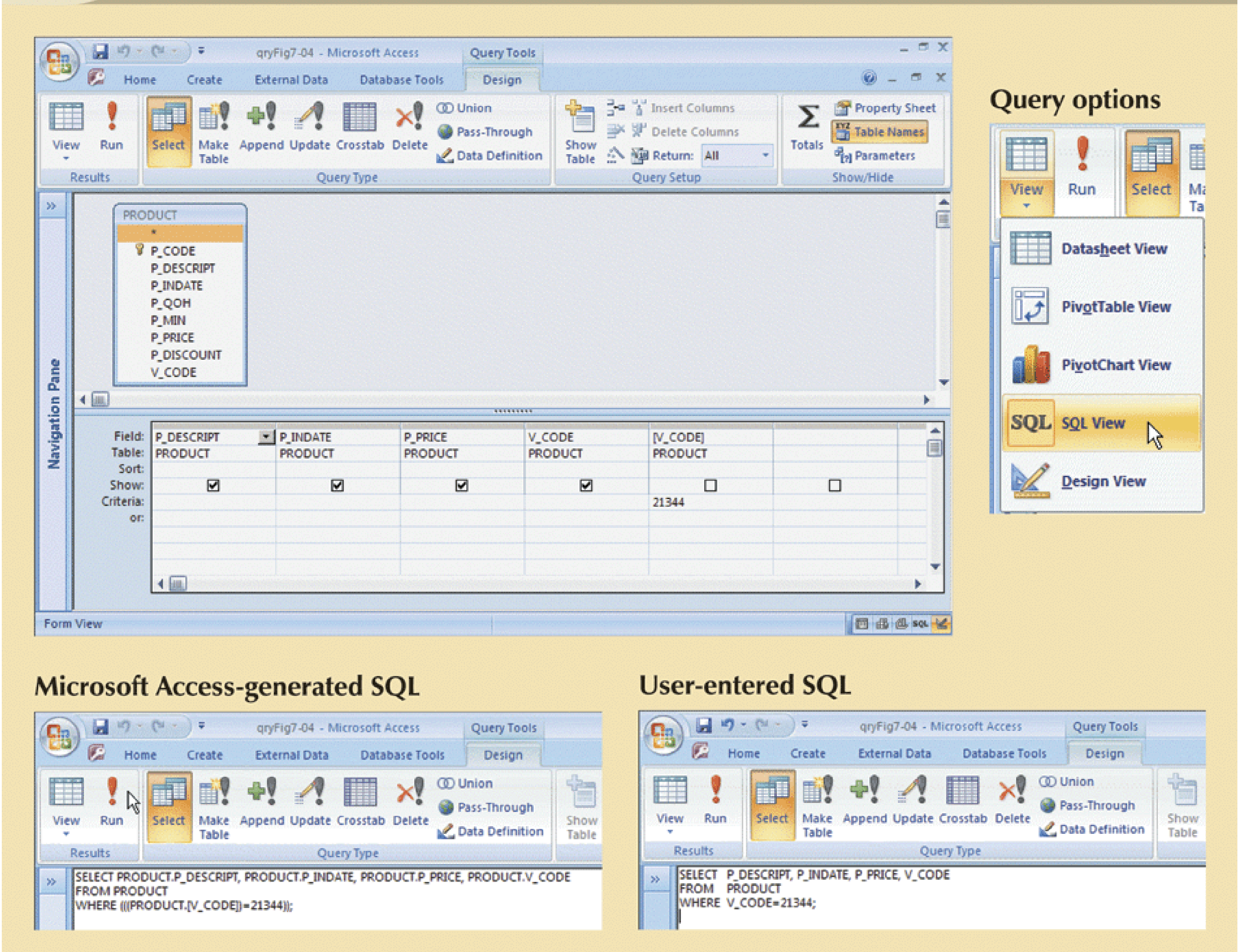
- Fine-tune SELECT command by adding restrictions to search criteria using:
  - Conditional restrictions
  - Arithmetic operators
  - Logical operators
  - Special operators

Selecting Rows with Conditional Restrictions

- Select partial table contents by placing restrictions on rows to be included in output
  - Add conditional restrictions to SELECT statement, using WHERE clause
- Syntax:

  SELECT *columnlist*
  FROM *tablelist*
  [WHERE *conditionlist*] ;

FIGURE 7.5 The Microsoft Access QBE and its SQL



Arithmetic Operators: The Rule of Precedence

- Perform operations within parentheses
- Perform power operations
- Perform multiplications and divisions
- Perform additions and subtractions

TABLE 7.7 The Arithmetic Operators

ARITHMETIC OPERATOR	DESCRIPTION
+	Add
-	Subtract
*	Multiply
/	Divide
^	Raise to the power of (some applications use ** instead of ^)

Logical Operators: AND, OR, and NOT

- Searching data involves multiple conditions
- Logical operators: AND, OR, and NOT
- Can be combined
  - Parentheses placed to enforce precedence order
    - Conditions in parentheses always executed first
- **Boolean algebra**: mathematical field dedicated to use of logical operators
- NOT negates result of conditional expression

Special Operators

- BETWEEN: checks whether attribute value is within a range
- IS NULL: checks whether attribute value is null
- LIKE: checks whether attribute value matches given string pattern
- IN: checks whether attribute value matches any value within a value list
- EXISTS: checks if subquery returns any rows

Advanced Data Definition Commands

- All changes in table structure are made by using ALTER command
- Three options
  - ADD adds a column
  - MODIFY changes column characteristics
  - DROP deletes a column
- Can also be used to:
  - Add table constraints
  - Remove table constraints

Changing a Column's Data Type

- ALTER can be used to change data type
- Some RDBMSs do not permit changes to data types unless column is empty

Changing a Column's Data Characteristics

- Use ALTER to change data characteristics
- Changes in column's characteristics permitted if changes do not alter the existing data type

Adding a Column
Dropping a Column

- Use ALTER to add column
  - Do not include the NOT NULL clause for new column
- Use ALTER to drop column
  - Some RDBMSs impose restrictions on the deletion of an attribute

Advanced Data Updates

- UPDATE command updates only data in existing rows
- If relationship between entries and existing columns, can assign values to slots
- Arithmetic operators useful in data updates
- In Oracle, ROLLBACK command undoes changes made by last two UPDATE statements

FIGURE 7.15 The cumulative effect of the multiple updates in the PRODUCT table (Oracle)

```
Oracle SQL*Plus
File  Edit  Search  Options  Help
SQL> SELECT P_CODE, P_DESCRIPT, P_INDATE, P_PRICE, P_SALECODE FROM PRODUCT;

P_CODE     P_DESCRIPT                          P_INDATE   P_PRICE P_SALECODE
---------- ----------------------------------- --------- -------- ----------
11QER/31   Power painter, 15 psi., 3-nozzle    03-NOV-07   109.99 2
13-Q2/P2   7.25-in. pwr. saw blade             13-DEC-07    14.99 2
14-Q1/L3   9.00-in. pwr. saw blade             13-NOV-07    17.49 2
1546-QQ2   Hrd. cloth, 1/4-in., 2x50           15-JAN-08    39.95 2
1558-QW1   Hrd. cloth, 1/2-in., 3x50           15-JAN-08    43.99
2232/QTY   B&D jigsaw, 12-in. blade            30-DEC-07   109.92 1
2232/QWE   B&D jigsaw, 8-in. blade             24-DEC-07    99.87 2
2238/QPD   B&D cordless drill, 1/2-in.         20-JAN-08    38.95 1
23109-HB   Claw hammer                         20-JAN-08     9.95 1
23114-AA   Sledge hammer, 12 lb.               02-JAN-08    14.40
54778-2T   Rat-tail file, 1/8-in. fine         15-DEC-07     4.99 2
89-WRE-Q   Hicut chain saw, 16 in.             07-FEB-08   256.99 1
PVC23DRT   PVC pipe, 3.5-in., 8-ft             20-FEB-08     5.87
SM-18277   1.25-in. metal screw, 25            01-MAR-08     6.99
SW-23116   2.5-in. wd. screw, 50               24-FEB-08     8.45
WR3/TT3    Steel matting, 4'x8'x1/6", .5" mesh 17-JAN-08   119.95 1

16 rows selected.

SQL>
```

Copying Parts of Tables

- SQL permits copying contents of selected table columns
  - Data need not be reentered manually into newly created table(s)
- First create the table structure
- Next add rows to new table using table rows from another table

FIGURE 7.16 PART table attributes copied from the PRODUCT table

PART_CODE	PART_DESCRIPT	PART_PRICE	V_CODE
11QER/31	Power painter, 15 psi., 3-nozzle	109.99	25595
13-Q2/P2	7.25-in. pwr. saw blade	14.99	21344
14-Q1/L3	9.00-in. pwr. saw blade	17.49	21344
1546-QQ2	Hrd. cloth, 1/4-in., 2x50	39.95	23119
1558-QW1	Hrd. cloth, 1/2-in., 3x50	43.99	23119
2232/QTY	B&D jigsaw, 12-in. blade	109.92	24288
2232/QWE	B&D jigsaw, 8-in. blade	99.87	24288
2238/QPD	B&D cordless drill, 1/2-in.	38.95	25595
23109-HB	Claw hammer	9.95	21225
23114-AA	Sledge hammer, 12 lb.	14.4	
54778-2T	Rat-tail file, 1/8-in. fine	4.99	21344
89-WRE-Q	Hicut chain saw, 16 in.	256.99	24288
PVC23DRT	PVC pipe, 3.5-in., 8-ft	5.87	
SM-18277	1.25-in. metal screw, 25	6.99	21225
SW-23116	2.5-in. wd. screw, 50	8.45	21231
WR3/TT3	Steel matting, 4'x8'x1/6", .5" mesh	119.95	25595

Adding Primary and Foreign Key Designations

- When table is copied, integrity rules do not copy
  - Primary and foreign keys manually defined on new table
- User ALTER TABLE command
  - Syntax:

    ALTER TABLE *tablename* ADD PRIMARY KEY(*fieldname*);
  - For foreign key, use FOREIGN KEY in place of PRIMARY KEY

Deleting a Table from the Database

- DROP
  - Deletes table from database
  - Syntax:

    DROP TABLE *tablename*;
- Can drop a table only if it is not the "one" side of any relationship
  - Otherwise RDBMS generates an error message
  - Foreign key integrity violation

Advanced SELECT Queries

- Logical operators work well in the query environment
- SQL provides useful functions that:
  - Count
  - Find minimum and maximum values
  - Calculate averages, etc.
- SQL allows user to limit queries to:
  - Entries having no duplicates
  - Entries whose duplicates may be grouped

Ordering a Listing

- ORDER BY clause useful when listing order important
- Syntax:

  SELECT *columnlist*

  FROM *tablelist*

  [WHERE *conditionlist*]

  [ORDER BY *columnlist* [ASC | DESC]];
- Ascending order by default

Listing Unique Values

- DISTINCT clause produces list of only values that are different from one another
- Example:

```
SELECT DISTINCT V_CODE
FROM    PRODUCT;
```

- Access places nulls at the top of the list
  - Oracle places it at the bottom
  - Placement of nulls does not affect list contents

Aggregate Functions

- COUNT function tallies number of non-null values of an attribute
  - Takes one parameter: usually a column name
- MAX and MIN find highest (lowest) value in a table
  - Compute MAX value in inner query
  - Compare to each value returned by the query
- SUM computes total sum for any specified attribute
- AVG function format similar to MIN and MAX

Grouping Data

- Frequency distributions created by GROUP BY clause within SELECT statement
- Syntax:

SELECT	*columnlist*
FROM	*tablelist*
[WHERE	*conditionlist*]
[GROUP BY	*columnlist*]
[HAVING	*conditionlist*]
[ORDER BY	*columnlist* [ASC | DESC]] ;

FIGURE 7.26 Incorrect and correct use of the GROUP BY clause

Oracle SQL*Plus

File Edit Search Options Help

```
SQL> SELECT V_CODE, P_CODE, P_DESCRIPT, P_PRICE
  2  FROM PRODUCT
  3  GROUP BY V_CODE;
SELECT V_CODE, P_CODE, P_DESCRIPT, P_PRICE
               *
ERROR at line 1:
ORA-00979: not a GROUP BY expression


SQL> SELECT V_CODE, COUNT(DISTINCT (P_CODE))
  2  FROM PRODUCT
  3  GROUP BY V_CODE;

    V_CODE COUNT(DISTINCT(P_CODE))
---------- -----------------------
     21225                       2
     21231                       1
     21344                       3
     23119                       2
     24288                       3
     25595                       3
                                 2

7 rows selected.

SQL> |
```

Virtual Tables: Creating a View

- **View** is virtual table based on SELECT query
- Create view by using CREATE VIEW command
- Special characteristics of relational view:
  - Name of view can be used anywhere a table name is expected
  - View dynamically updated
  - Restricts users to only specified columns and rows
  - Views may be used as basis for reports

Joining Database Tables

- Joining tables is the most important distinction between relational database and other DBs
- Join is performed when data are retrieved from more than one table at a time
  - Equality comparison between foreign key and primary key of related tables
- Join tables by listing tables in FROM clause of SELECT statement
  - DBMS creates Cartesian product of every table

Joining Tables with an Alias

- Alias identifies the source table from which data are taken
- Alias can be used to identify source table
- Any legal table name can be used as alias
- Add alias after table name in FROM clause
  - FROM *tablename alias*

Recursive Joins
Outer Joins

- Alias especially useful when a table must be joined to itself
  - Recursive query
  - Use aliases to differentiate the table from itself
- Two types of outer join
  - Left outer join
  - Right outer join

Summary

- SQL commands can be divided into two overall categories:
  - Data definition language commands
  - Data manipulation language commands
- The ANSI standard data types are supported by all RDBMS vendors in different ways
- Basic data definition commands allow you to create tables, indexes, and views

Summary (continued)

- DML commands allow you to add, modify, and delete rows from tables
- The basic DML commands:
  - SELECT, INSERT, UPDATE, DELETE, COMMIT, and ROLLBACK
- SELECT statement is main data retrieval command in SQL

Summary (continued)

- WHERE clause can be used with SELECT, UPDATE, and DELETE statements
- Aggregate functions
  - Special functions that perform arithmetic computations over a set of rows
- ORDER BY clause
  - Used to sort output of SELECT statement
  - Can sort by one or more columns
  - Ascending or descending order

Summary (continued)

- Join output of multiple tables with SELECT statement
  - Join performed every time you specify two or more tables in FROM clause
  - If no join condition specified, DBMX performs Cartesian product
- Natural join uses join condition to match only rows with equal values in specified columns
- Right outer join and left outer join select rows with no matching values in other related table